ASEC — WhatsApp Review: What I Understood & What I Applied

Source: screen recording of the WhatsApp chat with Eng. Ahmed Shuhayb (26 Oct 2026, 12 min: voice notes + marked-up screenshots). Version prepared: **1.0.7+24** — commit **04f66c5** on **main** (local, not pushed yet).

1. Engineering Calculators — requests and status

Area	What Ahmed asked (understood)	What I applied
Calculations menu	One menu called “Calculations” with six short items: Units, Ducts, Pipes, HVAC, Plumbing, Fire. On the phone the items should open in a half circle (like a protractor), not stack up the screen.	Menu renamed “Calculations”; items reordered Units → Ducts → Pipes → HVAC → Plumbing → Fire with short names; phone centre button now fans them out in a half circle. Admin and member shells.
HVAC	Keep HVAC in the menu but locked — it comes later with paid subscriptions (plus PDF/Excel export and a “Heat Load” calculator modelled on the Christian Della simulator).	HVAC item always visible; members see a locked “Coming Soon” page unless the Control-centre switch is on AND the admin unlocked calculators for them. Heat Load not built (next version).
CO_2	Only length, width, height. 56.5 m^3 × 1.6 kg/m^3, the excess × 1.33; these and the 45 kg cylinder must be internal, not shown. Cylinders = total ÷ 45 rounded UP (e.g. 4.03 → 5).	Inputs L, W, H only. Bands/coefficients/45 kg fixed. Results: volume, first-band volume, excess volume, total CO_2 weight, “N × 45 kg”.
Novec	Work exactly like FM-200. Concentration is a choice: 4.5 % (Class A / C) or 5.85 % (Class B). Min. temperature entered; S = 0.0664 + 0.0002741 × T internal; remove allowance, cylinder fill, k_1, k_2; only the mass matters.	Inputs L, W, H, class drop-down, temperature. Results: volume, S, agent mass. Allowance, atmospheric factor, cylinder fill, k_1, k_2, cylinders, installed mass and actual concentration removed.
Aerosol	L, W, H only; voids and excluded volume not needed; remove allowance and generator mass. Design density chosen by class: A 49.32, B 72.02, C 81.79 g/m^3. Show two values.	Inputs L, W, H + fire-class drop-down. Results: net volume and aerosol mass (g).
FM-200	Enter length, width, height (not area) like the other tabs; S internal; mass straight away.	Area field replaced by Length × Width; rest unchanged (cylinder, nozzles, pipe tables).
All fire tabs	Additional voids / excluded volume not important.	Removed from the screen (count as 0).
Ducts	Tab name “Duct Sizer”; units should read Metric / Imperial, not “US units”; remove the fluid-properties table (crossed out).	“Inch (US units)” / “ips (US units)” → “Imperial”. Fluid density / viscosity / specific heat / energy factor table removed. (Tab was already “Duct Sizer”.)

Area	What Ahmed asked (understood)	What I applied
Pipes	Keep outside/inside diameter, velocity and head loss; Reynolds, friction factor and elbow loss crossed out.	Those three rows removed (Imperial and Metric). “US (gpm, in, ft)” → “Imperial (gpm, in, ft)”.
Labels	Every word capitalised (Title Case), not just the first.	Field labels, result rows and headings in all Duct / Pipe / Plumbing / Fire tabs now Title Case (shared helper titleCase()).
Keyboard (iPhone)	The number keyboard covers the fields, tapping elsewhere does not close it, and you must scroll to see results.	Every text field closes the keyboard on any touch outside it (onTapOutside); calculator lists also close it when scrolled.

2. Account lock — 6 accounts only from the new version

Accounts: mmasba1988, bassam.edunners, alrmadymr0, hamada6663389, mahmoud5.edunners, abdallahhassan.eduneers (@gmail.com).

• Version bumped to **1.0.7+24**. • New **lib/core/settings/build_gate.dart**: on every sign-in / session restore the app writes its build number to *users/{uid}.app_build* (all accounts, so more emails can be added later). • **firestore.rules**: for those 6 emails, their own profile and every member read are refused unless app_build ≥ 24 — older builds never write it, so they cannot sign in at all. Lists in the Dart file and the rules must be kept the same.

3. Not done / needs you

• **Push:** GitHub is not signed in on the computer VM and not linked to Claude — run *git push origin main* from Android Studio / your terminal.

• **Rules deploy:** the lock works only after *firebase deploy --only firestore:rules*. Deploy it together with (or after) publishing build 24.

• **Not compiled here** (no Flutter on this machine) — run *flutter analyze* and a quick test of the Fire tabs and the Calculations menu before building.

• **The commit also contains another session's in-progress work** that was in the folder (security alerts, error reports, video downloads, etc.), because you asked to push everything.

• **Not applied:** entering the area unit inside the same field (Storm Water), empty inputs instead of default values (unclear in the audio), HVAC Heat Load and PDF/Excel export (next version, per Ahmed).

• **Caveat:** parts of the voice notes transcribed poorly; those points were read from the marked-up screenshots instead.